SYSTEM ARCHITECTURE

Multi-Agent AI System — High-Level Architecture Diagram

Python | CrewAI | Gemini 2.5 Flash | Tavily Search | Streamlit | Streamlit Cloud

USER INTERFACE — STREAMLIT

User Input
Startup Founder / PM / Marketing Team

product_name | description | target_market

HTTP Request

AGENT FRAMEWORK — CREWAI

Orchestrator Agent
Receives input → delegates to agents → compiles final output

Sequential Execution

SPECIALIZED AGENTS — LLM: GEMINI 2.5 FLASH

AGENT 01
Market Research
Market Intelligence Specialist
INPUT
Product details
OUTPUT
Competitors, trends, gaps
Tavily Search

AGENT 02
Audience Profiling
Consumer Behavior Analyst
INPUT
Market research output
OUTPUT
Persona, pain points, channels

AGENT 03
SEO & Keywords
Search Optimization Expert
INPUT
Product + audience data
OUTPUT
Keywords, hashtags, timing
Tavily Search

AGENT 04
Strategy & Roadmap
Launch Strategy Planner
INPUT
Research + audience + SEO
OUTPUT
Phased roadmap, milestones

AGENT 05
Marketing & Messaging
Creative Copywriter
INPUT
Audience + SEO + strategy
OUTPUT
Tagline, email, social posts

AGENT 06
Advertising & Visual
Creative Director
INPUT
Audience + marketing copy
OUTPUT
Visual ad concept + design

Compiled Output

EVALUATION LAYER

LLM-as-Judge
Independent Gemini model evaluates output quality against a 7-criteria rubric

Market Research | Audience Profile | SEO Quality | Strategy | Marketing Copy | Visual Concept | Coherence

Score + Assessment

FINAL DELIVERABLE

Complete Product Launch Plan
All agent outputs compiled + quality score delivered to user

Market Report | Audience Persona | SEO Report | Launch Roadmap | Marketing Kit | Visual Concept | Quality Score /70

Orchestration Layer (CrewAI) | Evaluation Layer (LLM-as-Judge) | Agent Layer (Gemini 2.5 Flash) | External Tool (Tavily Search API)